

DATA ANALYST PORTFOLIO · SOLOMON SHASANMI

Catching the 1 in 579: finding credit card fraud

THE QUESTION

If only 1 out of every 579 card transactions is fraud, how do you catch it without bugging thousands of honest customers, and is it worth the money?

PART 1 · THE RESEARCH

Why I did this, and what data I used

- **Fraud is a problem every bank and fintech has.** Credit unions, payment apps and online stores in the Northeast all hire analysts to watch transactions. I wanted to show I can do that job.
- **It's the hardest kind of problem to measure.** When something is this rare, the usual scorecards lie. Knowing which numbers to trust is the real skill here.
- **I used real data, not a made-up sample.** This is the classic public fraud dataset from a Belgian university (ULB) and the payments company Worldline. Anyone can download it and check my work.

The data at a glance

284,807

real card transactions

2 daysSeptember 2013, European
cardholders**492**

of them fraud (1 in 579)

28columns scrambled for
privacy

THE CATCH: MOST COLUMNS ARE SCRAMBLED

To protect cardholders, the bank blended the real details (things like merchant, location and device) into 28 mystery columns called V1 to V28. Think of a smoothie: you can taste that something's off, but you can't pick out the strawberry. So my model can tell *which* column raised the alarm, but not what it means in plain English. Only the amount and the time were left readable.

THE BOTTOM LINE

My best model caught **84 of 99** frauds it had never seen, with only **10 false alarms** out of almost 57,000 normal transactions. That saved about **two-thirds** of the money that would have been lost.

PART 1 · THE RESEARCH

How I did it, step by step

- 1 Pull the data and check it**
 A script downloads the data and confirms it's the right file: 284,807 rows and 492 frauds.
Why: If the data's wrong, everything after it is wrong.
- 2 Split it three ways**
 60% to learn from, 20% to practise on, 20% locked away for the final exam.
Why: Like studying, taking practice quizzes, then a final you've never seen. Only the final counts.
- 3 Test four ways to handle rare fraud**
 Weighting fraud more, making fake fraud examples (SMOTE), or throwing away normal ones.
Why: I wanted to see what each fix really does instead of assuming.
- 4 Let four models compete**
 XGBoost, Random Forest, Logistic Regression, and Isolation Forest (which never sees the answers).
Why: Pick the winner by evidence, not habit.
- 5 Set the cutoff by cost**
 Try every sensitivity setting on the practice set and keep the cheapest one.
Why: The right setting depends on money, not on a default.
- 6 Double-check, then build Tableau**
 A separate script redid all the math. Then I built 7 interactive dashboards.
Why: My check caught two of my own mistakes before they reached a chart.

Why I didn't grade it on accuracy

THINK OF IT LIKE

A weather forecaster who says "no tornado today" every single day is right 99.9% of the time, and completely useless. Same here: a model that calls **nothing** fraud is **99.83% accurate** and catches zero fraud. So I used better scorecards.

Four words you'll see on the next pages

Recall

Of all the real fraud, how much did I catch?

Like: of all the fish in the lake, how many ended up in my net.

Precision

Of everything I flagged, how much was really fraud?

Like: how much of my net is fish, and how much is old boots.

Cutoff

The score where "fine" becomes "flag it."

Like: a smoke alarm's sensitivity dial.

False alarm

An honest purchase that got flagged.

Like: the alarm going off because you made toast.

Money rules I used: a missed fraud costs its full amount; every flagged transaction costs €10 for a person to review. The €10 is my assumption, and the dashboard lets you slide it.

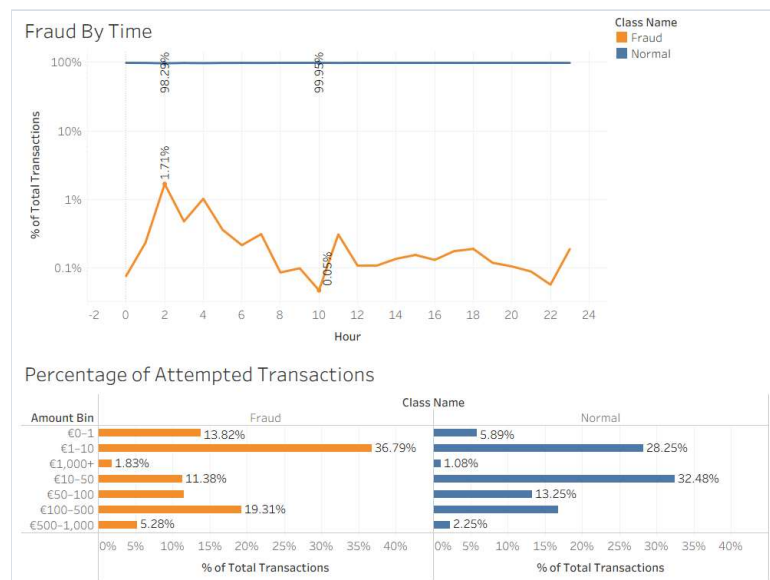
PART 2 · WHAT I FOUND

FINDING 1 · THE DATA

Fraud is rare, hides in small amounts, and likes the middle of the night

THE BIG IDEA

Only **1 in 579** transactions is fraud, but at 2am it jumps to about **1 in 58**, ten times the usual rate.



From my Tableau dashboard. The top chart uses a log scale, so each gridline is 10× the one below.

WHAT TO NOTICE

- Top chart: the orange fraud line spikes at 2am, when fewer people are shopping.
- Bottom chart: about half of all fraud is under €10, compared with a third of normal purchases. That fits a common trick: testing a stolen card with a tiny purchase first.

SO WHAT?

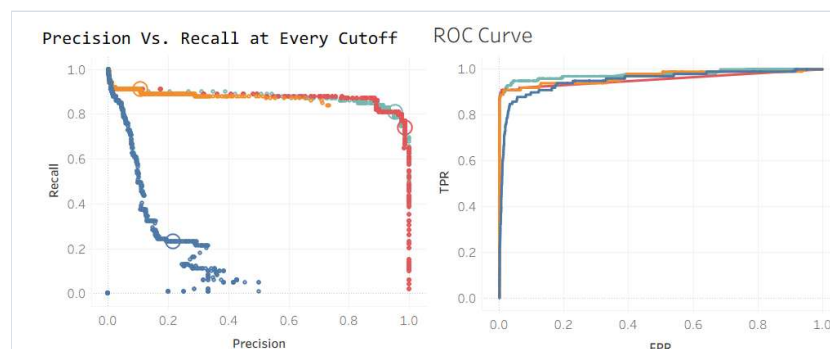
Watch small overnight purchases more closely. They match the common “is this card still working?” test.

FINDING 2 · THE MODELS

XGBoost wins, and the fix you'd expect isn't the one that matters

THE BIG IDEA

XGBoost scored best, with Random Forest a close second. The popular imbalance fixes barely helped.



Colours: teal XGBoost, red Random Forest, orange Logistic Regression, blue Isolation Forest.

WHAT TO NOTICE

- Left: the good models hug the top-right corner. XGBoost (teal) and Random Forest (red) do best; Isolation Forest (blue), which never saw the answers, falls apart.
- Right: on the ROC chart **every** model looks great, even the weak one. With rare events that chart flatters everyone, so I didn't trust it.

SO WHAT?

Making fake fraud examples (SMOTE) didn't make the model smarter. It just made it jumpier: precision fell from 79% to 6%. What really mattered was **where you set the cutoff**.

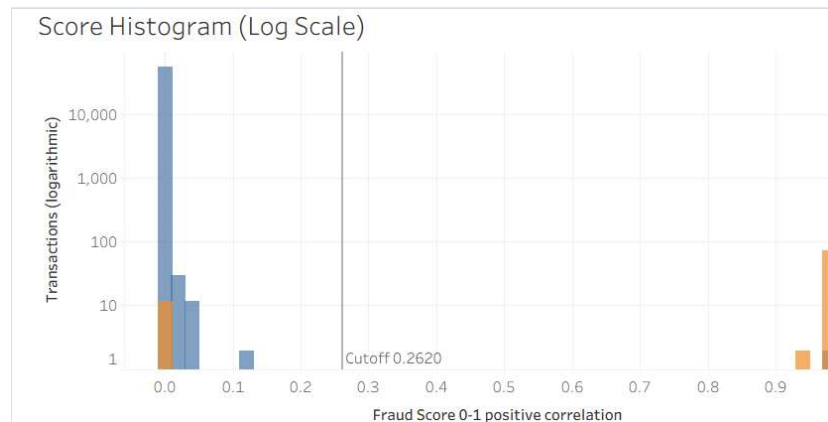
PART 2 · WHAT I FOUND

FINDING 3 · THE CUTOFF

Drawing the line: 84 caught, 10 false alarms

THE BIG IDEA

At the best setting, the model caught **84 of 99** frauds and bothered only **10** honest customers out of 56,863.



WHAT TO NOTICE

- Normal purchases (blue) pile up at the left, near a score of 0. Fraud (orange) piles up at the right, near 1.
- The grey line is the cutoff. Orange left of it = missed fraud (15). Blue right of it = false alarms (10).

From my Tableau dashboard at XGBoost, cutoff 0.262. On the live dashboard, dragging the cutoff updates the counts.

SO WHAT?

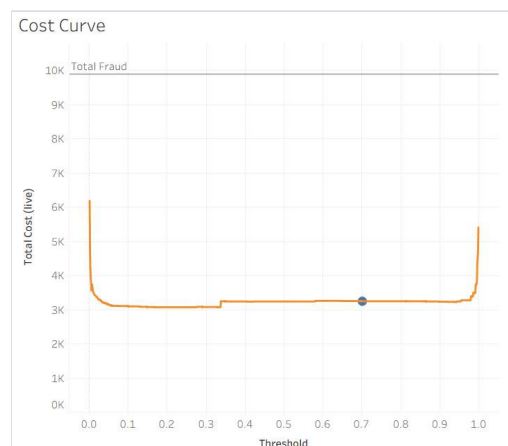
A clean split like this means a small review team can keep up: only 94 alerts out of almost 57,000 transactions, and 9 out of 10 are real fraud.

FINDING 4 · THE MONEY

What it's worth: two-thirds of the loss, saved

THE BIG IDEA

Out of **€9,874** of fraud in the test set, the model saved about **€6,426 (65%)** after paying for reviews.



WHAT TO NOTICE

- The flat grey line is doing nothing: every fraud gets through.
- The orange curve is the model's total cost. Too sensitive (far left) and review costs explode; too relaxed (far right) and fraud slips by. The middle is a wide, flat valley.

From my Tableau dashboard (shown with review cost set to €6; the report's numbers use €10).

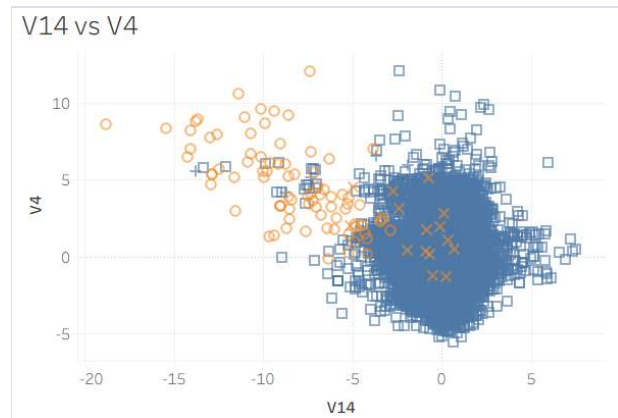
SO WHAT?

Because the valley is wide, the business has room to choose. A bank that hates false alarms can slide right without losing much.

PART 2 · WHAT I FOUND

FINDING 5 · THE REASONS

Two hidden columns separate most fraud from normal



Orange circles = fraud, blue squares = normal, x = missed fraud.

Plot every transaction by just two scrambled columns, V14 and V4, and most fraud (orange) sits off by itself, away from the big blue cloud. V14 was the top reason behind **84 of the 94** alerts.

The frauds the model missed (x) are hiding **inside** the blue cloud. They look just like normal purchases, which is why no model catches everything.

PART 3 · PUTTING IT TO WORK

Who uses this, and why it matters

Banks & credit unions

"Which card transactions should we hold for review?"

Exactly this project, on their own data, with real reasons like "new device" instead of V14.

Payment apps & fintech

"How do we stop fraud without annoying good customers?"

The cutoff slider shows the trade-off between blocked fraud and blocked customers.

Online stores

"Which orders are likely chargebacks?"

Small test purchases at odd hours are the same warning sign.

Insurance

"Which claims deserve a second look?"

Only a few claims are fake, so it's the same rare-event problem and the same scorecards.

Hospitals & health insurers

"Which bills look like billing fraud?"

Rank claims by risk so auditors check the likeliest ones first.

Colleges

"Are some 'students' just collecting financial aid?"

"Ghost student" sign-ups are rare and costly: the same approach, different data.

What I bring

- **I pick the right scorecard.** I didn't get fooled by 99.8% accuracy, and I measured results in euros, which is what a manager actually cares about.
- **I check my own work.** My verify script caught two of my own bugs, and a data download that was silently missing a column, before any of it reached a dashboard.
- **I let the business make the call.** The dashboard has sliders for model, cutoff and review cost, so a manager can see the trade-off instead of taking my word for it.

Limits: two days of 2013 data; 28 scrambled columns; only 99 frauds in the final test, so the exact numbers could shift with more data; the €10 review cost is an assumption. Data: ULB Machine Learning Group & Worldline, via OpenML (dataset 42175). Charts are exported from my Tableau workbook; numbers are checked by `verify.py`.